Measuring Agreement in the Ramban Supercommentary Tradition

A knowledge-graph distance analysis of 21 commentators across 38 sections

Final report · 2026-09-06 · updated 2026-10 · all analyses reproducible from `RambanKGAnalysis/scripts/`

> **Correction, October 2026.** A check against the source (Eliezer Baumgarten) found a labelling error in the Da'at Chacham units. `DaatChachamVatokan114` is Da'at Chacham, **Vat. 214** (there is no Vat. 114 witness). `DaatChachamVatokan214` is the **marginal notes on Da'at Chacham, Vat. 214**, a separate unit, not a witness. Texts, graphs and all numbers are unchanged; identifiers are file names and are kept. Readings affected are marked below.

Executive summary

We built knowledge graphs for 21 supercommentators on Nachmanides (Ramban) across 38 sections of his Torah commentary – 385 graphs in all – and measured how far apart they are, using a five-rung distance ladder over four different notions of "the same concept".

The headline is a negative one, and it is robust. The supercommentators do not form schools. What the corpus contains is *manuscript families* – witnesses of the same text – set against a broad and near-total dispersion. Commentators writing on the *same passage of Ramban* share about 8% of their concepts, and of the claims they could both have made about that shared 8% they agree on almost none.

Six results carry the report:

1. **No schools.** Cluster structure beats chance in 13 of 36 sections, but most of those are a single commentator separating from the rest (12+1, 9+1), not two camps. Balanced structure exists in only a handful (§18 at 5+5 is the clearest). Denser graphs show *no more* structure than sparse ones, so this is not an artefact of thin extraction.
2. **Divergence is total, and extends to the proof-texts.** Median locus overlap 0.080; median reading agreement 0.000. Even scriptural anchors recur across commentators only 35% of the time – less often than a divine name or a biblical figure.
3. **Manuscript families are recovered cleanly**, and three previously unremarked candidates emerge, each confirmed by two independent instruments.
4. **IbnShuib is the closest tracker of Ramban** on four measures – content, order, grouping and register.
5. **The distance ladder is not redundant**: 45 aligner×rung cells reduce to 29 independent axes.
6. **The measurement scale matters more than expected.** A raw rung score is *ordinal*; calibrated into bits of evidence, a gap between 0.9 and 0.7 turns out to be worth **zero** additional evidence.

1. What was measured, and how

1.1 The corpus

Graphs	385 (`data/graphs_canonical/`)
Sections of Ramban	38
Commentators	21 (12 with ≥10 sections)
Section × commentator grid	48.2% dense
Median graph	35 concepts / 41 edges

Two graphs are empty (`kg_7_2_RAMBAN`, `kg_9_UnKnownEskorial`) and one has two concepts (`kg_8_UnKnownEskorial`); all three are excluded from every ranking. Section §1′s RAMBAN graph is a correct graph of the *wrong passage* – a known data error, retained deliberately as a control.

1.2 The distance ladder

rung	question
D_1 content	do they discuss the same concepts?
D_2 structural (L_0/L_1/L_2)	do they make the same claims about them?
D_3 anchoring	do they root the same idea in the same scriptural place?
D_4 positional	do they place concepts in the same neighbourhood?
D_5 sequence	do they present ideas in the same order?

1.3 Four notions of identity

A distance is only meaningful once "the same concept" is defined. Four definitions were run as separate rows and compared:

- **Method I – embedding**: same *meaning*, by cosine above a threshold. The only method with an absolute scale, and therefore the only one that can say "no".
- **Method II – canonical authority**: same *referent*, decided by resolving a quotation to a canonical verse. Deterministic and corpus-global, but reaches only 12.9% of concepts.
- **I+II composite**: verse first, then embedding, then label. This turned out to be the best row on **every** rung against a negative control.
- **I∧III corroborated**: embedding detects, structural role corroborates a relaxed band. Method III is *never* used alone – a size-matched panel showed it matches ~18–19 nodes in *unrelated* pairs exactly as in related ones, so it has no "no".

1.4 The negative control

Every claim in this report is measured against **600 size-matched cross-section pairs** – commentators writing on *different* passages, therefore known to be unrelated. This is what converts a similarity score into evidence. Its availability was the decisive argument for keeping the unit of analysis at the section level rather than merging each commentator into one graph: per-section yields **72,024** known-unrelated pairs, per-commentator only **38**.

1.5 Measurement scale

A raw rung score is **ordinal** – higher means more similar, and nothing more. "Distance" promises more than we measure. Calibrated against the negative control, the same score becomes **bits of evidence** ($\log_2$ of the likelihood ratio), which has a true zero at "no evidence" and on which differences are meaningful.

> **The clearest illustration:** in the best cell, a raw score of **0.9 and a raw score of 0.7 carry the identical +6.33 bits** – an odds ratio of 1.0. Both land in the top calibration bin, because almost nothing in this corpus scores that high. The ordinal reading makes the gap look decisive; the interval reading shows it is empty.

A ratio scale of *closeness* is not available: it would need a unit of textual closeness, and there isn't one. A ratio scale of *evidence* is available, and is what we use.

1.6 A global vocabulary

Pairwise alignment mints a concept's key *per pair*, so the same concept has a different name in every comparison – which makes statements like "these four commentators share concept X" literally unsayable. We therefore built **multiple graph alignment** (`graphcompare/multi.py`): one global key space per section, validated to recover the known manuscript twins at **2.42×** separation from non-twins while still flagging the §1 data error.

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2. Findings by research question

RQ #1 – Measurement

The distance matrix reproduces the earlier E3 analysis at **ρ = 0.950** across 339 vs-Ramban pairs, confirming the instrument before any new claim rests on it. Two corpus defects surfaced in the process: the two empty graphs, and confirmation that §1 is a graph of the wrong passage.

The ladder is **not redundant**: across 1,873 pairs, 45 aligner×rung cells collapse to only **29 independent axes** at ρ ≥ 0.98. The rungs measure genuinely different things – a methodological result in its own right for graph-

distance work on humanities corpora.

A clean negative: switching Method I from directional `argmax` to symmetric `mutual` changes the pair ordering by **ρ = 0.996** – i.e. essentially nothing. The directional matching choice in the earlier analysis was not load-bearing.

RQ #2 – Explaining similarity

Across **1,534 horizontal pairs** (commentator ↔ commentator):

- The accepted band (central 80%) is a raw score of **0.076 – 0.354**.
- **18% of pairs carry zero or negative evidence** – no more similar than two commentators writing on unrelated passages.
- The threshold of "0.50" suggested in the source research document sits at the **96.8th percentile**: only 49 pairs of 1,534 reach it.

Outlier cutoffs were derived rather than chosen: *divergent* = bits ≤ 0 (straight from the null); *close* = at least as similar as the lower quartile of 42 confirmed manuscript-twin instances.

The close tail validates the instrument. `KeterShemTov ↔ KeterShemTovMS_L` tops it in four separate sections; `DaatChacham107 ↔ 114` (the two Da'at Chacham witnesses, Vat. 107 and Vat. 214) in two more.

⭐ **Three candidate relationships not in the known gold set**, each recurring across sections:

pair	sections
`UnKnownEskorial ↔ UnKnownVatikan214`	§19_2, §20, §26
`DaatChachamVatokan214` (marginal notes, Vat. 214) `↔ UnKnownEskorial`	§22, §25_2
`MeiratEinayim ↔ SaportaOxford1645`	§15

Recurrence is what makes these more than topical coincidence – one section could be shared subject matter; three cannot. **All three were later confirmed independently** by co-clustering (RQ #3).

✅ **The "214 anomaly" is resolved, and it was a detection.** The earlier analysis grouped DaatChacham 107, 114 and 214 as one family. Our measurements placed **214 outside** that group (107↔214 and 114↔214 rank at the 23rd–40th percentile, while 107↔114 ranks at the 99th) and attached it instead to `UnKnownEskorial`. The source check (October 2026) explains why: the unit labelled 214 is the marginal notes on Da'at Chacham, not a witness, while the unit labelled 114 is the real Vat. 214 witness. Without being told, the measurements separated the one unit that is not a Da'at Chacham witness.

RQ #3 – Clusters

Per section: cluster structure beats a permutation null in **13 of 36 sections** at p < 0.05 (5 survive Benjamini-Hochberg; 3 survive Bonferroni: §26, §30_1, §32).

⚠️ **But read the cluster sizes.** Most significant splits are a single commentator separating from the rest – §26 and §24 are 12+1, §22 is 9+1. Genuinely balanced structure appears in only a few: **§18 (5+5, p = 0.005)**, §32 (4+2+2), §5, §7_1. Silhouettes stay low throughout (0.05–0.32) and mean pairwise distance is 0.83–0.96. **The corpus has outliers more than it has camps.**

Across sections – "local or consistent?" For each pair of commentators we measured the fraction of shared sections in which they land in the same cluster:

pair	co-cluster rate	sections
KeterShemTov ↔ KeterShemTovMS_L	**1.00**	17
UnKnownEskorial ↔ UnKnownVatikan214	**1.00**	7
DaatChacham107 ↔ 114 (= Vat. 214)	**0.96**	25
Likutim_bhml_1978 ↔ UnKnownEskorial	0.80	5

The median over 105 pairs is **0.32 – chance**. So the only consistent groupings in this tradition are **manuscript families, not schools**.

Is the absence of schools real, or just sparse graphs? Tested directly: if sparsity were suppressing structure, denser sections would show more of it. They do not – concepts→silhouette ρ = −0.178, and −0.143 controlling for the number of commentators. Sparsest and densest halves have median silhouettes of 0.106 and 0.097.

🌟 And an unexpected finding in the same test: **more material makes commentators *more* distant, not less** (concepts → mean pairwise distance ρ = +0.227). They do not converge as they say more; they diverge further.

RQ #4 – Dominance

The plan proposed centrality on the similarity graph. We reframed, for two reasons the plan itself anticipates: dominance is *directional* and a symmetric distance cannot express direction; and a graph with mean distances of 0.83–0.96 whose significant structure is mostly lone outliers has no centre for centrality to find.

In a supercommentary tradition the centre is not an emergent hub – it is the base text. The question becomes: who stays closest to Ramban?

#	commentator	content	order	co-cluster	register	mean rank
1	**IbnShuib**	0.345	0.914	0.69	0.971	**11.8**
2	UnKnownEskorial	0.351	0.874	0.54	0.945	11.0
3	Ricanati	0.286	0.863	0.45	0.940	9.8
…						
12	Comment_a_shuib_Parma_68	0.109	0.405	0.36	0.881	2.0

IbnShuib leads on all four measures. The commentator at the bottom is the same one that dominates the divergent tail in RQ #2 and forms the lone-outlier split in RQ #6.

⚠ Two qualifications. The four measures are **not independent** – content and order correlate at +0.91 – so this is a *stable ranking*, not four independent confirmations. And **closeness is not influence**: this says who tracks the base text, not who shaped the tradition, which would require citation direction and chronology that the graphs do not carry.

RQ #5 – Characterising divergence

We decomposed each pair into **locus overlap** (do they engage the same material?) and **reading agreement** (of the claims they could *both* have made about their *shared* concepts, how many do they share?). The conditional denominator is what separates "talking about different things" from "reading the same thing differently".

- median locus overlap **0.080**
- median reading agreement **0.000**
- **27%** of pairs share concepts but never link the same two – no comparable claim exists at all
- of those that do have comparable claims, **59% agree on nothing**

⚠ **Both measures sit at the floor, and they are not alternatives.** It would be wrong to report "divergence is about reading rather than locus": an 8% locus overlap means they mostly do not meet on the same ideas in the first place. **Divergence is total.**

Which kinds of concept are shared (share rate = appears in ≥2 commentators of the same section):

class	share rate
SpiritualEntity	0.391
BiblicalFigure	0.390
PhysicalObject	0.371
scriptural anchor (verse)	**0.349**
TheologicalConcept	0.320
KabbalisticConcept	0.306
BiblicalSource (a book)	0.192
OtherConcept	0.168

The specific question posed was *verses versus kabbalistic terms*. Verses are shared more – but only modestly, 0.349 against 0.306.

⚠️ **The "fixed point" intuition is only weakly supported.** If every commentator had to pass through the same verses, the anchor rate should tower over the concept types. It sits mid-table instead: they are **less likely to share a verse than to share a divine name or a biblical figure**.

⭐ **The asymmetry is the reportable claim.** Verses beat both *doctrinal* classes but lose to the *concrete* ones. What this tradition holds in common is its **cast and furniture** – who and what the passage is about – more than either its proof-texts or its doctrine.

Most widely shared anchors: **Gen 9:13** (13 commentators, §5) · **Hab 3:2** (11, §12) · **Lev 23:40** (10, §27).

RQ #6 – The nature of the vertices

Since the commentators do not group by content, do they group by **method** – the kinds of node and edge they favour? Distance here is the Jensen-Shannon divergence between type distributions.

⚠️ Ramban himself must be excluded: with him included the best split is trivially {Ramban} vs {everyone}, because the base text is not a kabbalistic supercommentary (`KabbalisticConcept` −15.9 percentage points).

Among the 12 supercommentators, **only the single-outlier split is significant**:

k	cluster sizes	silhouette	p
2	**11+1**	0.529	**0.000**
3	9+2+1	0.259	0.664
4	9+1+1+1	0.204	0.854
5	5+4+1+1+1	0.272	0.680
6	4+3+2+1+1+1	0.290	0.570

Every *balanced* partition fails. **They do not group by method either.**

But style is a genuinely independent axis – its correlation with content co-clustering is −0.166, near zero. The profiles measure something the content rungs do not; they simply do not partition the tradition. As descriptive signatures they work well:

- **IbnShuib** is the *least* kabbalistic supercommentator (`KabbalisticConcept` −8.9pp, scriptural quotation +3.9pp) – coherent with his being the closest tracker of Ramban.
- **KeterShemTovMS_L** is the most kabbalistic (+7.0pp).
- **Comment_a_shuib_Parma_68** is the outlier: `PhysicalObject` +7.2pp, `BiblicalQuotation` −7.3pp.

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3. Cross-cutting findings

Overlap is not agreement. The corpus repeatedly produces pairs with substantial shared vocabulary and no shared claims – the diagnostic case being pairs at $D_1 = 0.364$ with $D_2 \cdot L_2 = 0.000$. Any similarity measure reported without its structural companion will overstate agreement.

The instrument has a floor that must be reported with the findings. Per-pair significance testing is mostly unavailable on this corpus: under the appropriate null, 96% of pairs are degenerate for $D_2 \cdot L_2$, 84% for D_3, 52% for D_4. Aggregate evidence (bits over the corpus) survives; per-pair significance largely does not. This is a property of the corpus, not a failure of the run, and it explains why claims requiring per-pair significance could not be substantiated.

Evidence is cheap; interpretation is not. The entire computational campaign cost **$0 in LLM spend** except one verification exercise at **$0.25**. Every substantive limitation encountered was methodological, not budgetary.

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4. Corrections and limitations

We record the errors found and fixed during the campaign, because two of them changed conclusions.

🛑 **A broken permutation null (found 2026-09-06) invalidated the first version of RQ #3.** The null permuted rows and columns of the distance matrix by the *same* permutation – a pure relabelling that leaves the geometry untouched, so the clustering and its silhouette were identical in every draw (verified: 50 draws, all 0.908940). The p-values it produced were tie-breaking noise. Corrected, the result moved from "1 of 36 sections" to **13 of 36**. The corrected reading is in RQ #3.

🛑 **A confounded measure closed Step 6 (the vertical lens) as a negative.** Comparing whole systems by repertoire overlap turned out to be ~76% explained by how much a commentator wrote (ρ = +0.874 between repertoire size and vertical closeness). The apparent "drift" signal was coverage, not divergence. Whole-system comparison needs a size-controlled statistic; the raw version is not reportable.

⚠️ **A stored methodological claim was amended.** An earlier finding held that structural corroboration at a relaxed threshold gave a 4 : 1 precision advantage. Re-measured on two corpora, it was *below parity* at the original parameters, and only becomes load-bearing after moving the threshold from 0.55 to 0.68. The qualitative claim survives; the numbers were withdrawn.

⚠️ **$D_3 = 1.0$ means "no grounding evidence", not "perfect agreement"** – the top disagreements between D_1 and D_3 are pairs with zero shared content and empty grounding sets on both sides.

⚠️ **D_5's local-best score saturates on size-mismatched pairs**, and aggregated per commentator it is largely a restatement of content overlap (ρ = 0.865). Sequence claims must be made pair-wise, or with a length-controlled statistic.

Known limitations of scope:

- The unit of analysis is the section. The candidate pair `UnKnownVatikan214 ≈ DaatChachamVatokan214` (the anonymous unit and the marginal notes, both in Vat. 214) shares no section and is therefore unmeasurable here – an accepted cost of the decision. Whether the anonymous unit is part of the marginal notes is an open question for the scholars.
- Only the `verified` tier of scriptural anchoring carries a canonical verse (2,454 of 3,499 quotations, 70%), so loose citation is undercounted and the verse share rate of 0.349 is a **lower bound**.
- The global vocabulary is built per section, not per corpus. Statements comparing concept identity *across* sections are outside what this analysis supports.
- An estimated **~10.4% of alignment merges are wrong** (27.9% in the 0.80–0.85 similarity band, 4.0% above it), measured by LLM adjudication of 1,812 merges.

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5. What a scholar should look at

1. **The three candidate manuscript relationships** in RQ #2, each confirmed by two independent instruments, with their shared concepts, scriptural anchors and shared claims in `data/step3/3_similarity_band.md`.
2. **The 214 anomaly** – a direct challenge to an existing grouping.

3. **§18**, the one section with a clean balanced two-camp split (5+5, p = 0.005).
4. **The divergent pairs** in `data/step3/`, each with the claims each side makes *instead* – for example §26, where one witness argues in a numerological-sefirotic register (עשרה → כח אחד, חפ״ת → ג׳ אבות) while another argues in a *midat ha-din* register (דין בדין → דין קשה/רפה, עלוי העטרה → תרועת מלחמה). Same passage of Ramban, different conceptual worlds.

Appendix – artifacts

analysis	outputs	script
E3 reproduction	`data/step2a/`	`step2a_methodI.py`, `step2a_correlate.py`
Aligner × rung matrix	`data/step2b/matrix.csv`, `2b_results.md`	`step2b_matrix.py`, `step2b_analyze.py`
Gold bake-off, redundancy, D5-vs-Ramban	`data/step2b/2c_2e_results.md`	`step2c_2e.py`
Measurement scale (bits)	`data/step2b/2f_scales_results.md`	`step2f_scales.py`
Null testability	`data/step2b/2g_null_testability.md`	`step2g_nulls.py`
Similarity band + outliers (RQ #2)	`data/step3/`	`step3_band.py`
Global alignment + acceptance	`data/mga/`	`mga_validate.py`, `mga_verify_sample.py`
Clusters (RQ #3)	`data/step4/`	`step4_clusters.py`, `step4b_density_check.py`
Divergence (RQ #5)	`data/step5/`	`step5_divergence.py`, `step5b_verses.py`
Vertex profiles (RQ #6)	`data/step6/6_vertex_profiles.md`	`step6_vertex_profiles.py`
Vertical lens (Step 6)	`data/step6/6_vertical_lens.md`	`step6_vertical_lens.py`
Dominance (RQ #4)	`data/rq4/`	`rq4_dominance.py`

Framework code: `simplekg/graphcompare/` – `multi.py` (multiple alignment), `scales.py` (measurement scale), `nulls.py` (significance), `corroborate.py`, `role.py`, `compare.py`, `reduce.py`.